SUPPLIER RELIABILITY MATRIX

Project Development Document

A Personal Prototype in Supply Chain Risk Analytics

Prepared by

Shubham Gambhire

Prototype status · Not affiliated with, and containing no data from, any employer

Contents

1. Executive Summary 3
2. Background and Origin 3
3. Objectives 3
4. Scope and Design Philosophy 3
5. Scoring Methodology 4
6. System Architecture and Technology Stack 6
7. Feature Overview 6
8. Design and User-Experience Evolution 7
9. Data Privacy and Confidentiality Approach 8
10. Testing and Validation Approach 8
11. Known Limitations and Future Work 8
12. Conclusion 9

1. Executive Summary

The Supplier Reliability Matrix is a self-contained, browser-based tool that scores suppliers on delivery reliability and sourcing risk using only the data a real procurement team can be expected to maintain. It began as a scoring model built during an internship to support supplier risk reviews, and has been rebuilt here as an interactive prototype: an editable supplier ledger, a live risk-matrix visualization, a fully tunable scoring engine, and a documented methodology that states its own limitations rather than hiding them.

This document records why the tool is built the way it is, how the scoring methodology works end to end, the technical approach behind it, and what was deliberately left out and why. It is written as a project record, not marketing copy: known gaps are named alongside what shipped.

2. Background and Origin

The scoring logic underneath this tool originates from a Supplier Risk Scorecard built in Excel during an internship, used to flag suppliers worth a closer look ahead of a sourcing or renewal decision. That original workbook scored suppliers on a small set of operational fields, on-time delivery history, sole-source status, and an internally assigned priority tier, because those were the fields the business actually kept current. Everything else a formal supplier-risk model might ask for, quality rejection rates, invoice accuracy, financial health, simply was not reliable enough to use.

This project reimplements that logic as a standalone web application. All company names, supplier names, and locations used in the example data are entirely fictional (see Section 9). No data, real or derived, from the original employer or its suppliers appears anywhere in this tool.

It is one of a small portfolio of independent prototypes built to apply and demonstrate supply chain domain knowledge, aimed at digital supply chain and supply chain analytics roles.

3. Objectives

- Reimplement a real, spreadsheet-based supplier scoring methodology as an interactive, explorable tool rather than a static report.
- Demonstrate the ability to make a deliberate methodological simplification and defend it on its merits, not just build a working interface.

- Produce something closer to a working internal tool than a portfolio mockup: editable data, persistence, validation, and a documented methodology a stakeholder could actually question.
- Arrive at a visual identity distinct from generic, template-driven dashboard design, and grounded in the subject matter rather than borrowed from an unrelated prior project.

4. Scope and Design Philosophy

4.1 Why Six Fields, Not Sixteen

A textbook supplier-scoring model reaches for a long list of KPIs: lead-time variance, quality rejection rate, invoice accuracy, responsiveness, financial health, and more. In practice, most procurement teams do not keep all of that clean, and a model that depends on data nobody maintains does not get used. This tool scores only what a standard purchase-order ledger and vendor master file reliably contain: delivery counts, a priority tier set by the category manager, and whether a supplier is sole-sourced. Everything else is a reasonable next iteration once that data exists, not an oversight in this version.

4.2 Why Not a Full Multi-Criteria Decision-Making (MCDM) Method

Structured multi-criteria methods, AHP, DEMATEL, TOPSIS, and similar, are the standard academic and consulting toolkit for supplier evaluation. They were deliberately not used here, not because they are inferior, but because they solve a different problem than this tool does.

	AHP / DEMATEL / TOPSIS	This Tool
Typical use	Periodic supplier selection or criteria weighting	Continuous monitoring of suppliers already on file
Input process	Expert panel, pairwise comparison matrices	Transactional data already in the PO ledger
Cadence	Deliberative; annual review or RFP cycle	Automated; re-runs on every data change
Handles thin data?	Assumes every criterion has a trustworthy score	Explicit Insufficient Data guard (Section 5.6)
Models criteria interaction	Yes (DEMATEL explicitly)	Partially, via a compounding penalty (Section 5.3)

A two-criterion weighted model was judged sufficient for this tool's scope; the honest gap this leaves is addressed directly in Section 10.

5. Scoring Methodology

Every supplier is scored through five steps, plus an optional secondary comparison. Every weight, threshold, and multiplier below is user-editable in the application's Settings panel and recalculates the entire tool live.

5.1 Delivery Score

On-Time Rate is On-Time Deliveries divided by Total Purchase Orders, bucketed into a 1–5 score:

On-Time Rate	Delivery Score
≥ 90%	5
75–89%	4
60–74%	3
45–59%	2
< 45%	1

5.2 Dependency Score

Combines sole-source status with the supplier's assigned priority tier:

Condition	Score
Not sole-sourced (alternates exist)	5
Sole-sourced, Priority 4–5	3
Sole-sourced, Priority 3	2
Sole-sourced, Priority 1–2 (most critical)	1

5.3 Performance Score and the Compounding Penalty

Performance Score = (0.65 × Delivery Score) + (0.35 × Dependency Score) − Compounding Penalty.

Delivery Score and Dependency Score are measured independently, but the risk they represent is not. A sole-sourced supplier that is also missing delivery dates is worse than the sum of those two problems, because there is no fallback to absorb the miss. The compounding penalty activates only when both scores sit below the midpoint of the 1–5 scale, so a supplier weak on just one dimension is not double-penalized:

Compounding Penalty = weight × max(0, 3 − Delivery Score) × max(0, 3 − Dependency Score), weight defaults to 0.15 and can be set to 0 to fall back to a plain weighted sum.

5.4 Priority Multiplier and Risk Score

Priority Tier	Multiplier
P1 (most critical)	1.5x
P2	1.3x
P3	1.1x
P4	1.0x
P5	0.9x

Risk Score = Performance Score × Priority Multiplier.

5.5 Risk Tier

Risk Score	Tier
≥ 3.75	Low Risk
2.50–3.74	Medium Risk
1.50–2.49	High Risk
< 1.50	Critical Risk

5.6 Insufficient Data Guard

Suppliers with fewer than three total purchase orders on file are flagged Insufficient Data rather than scored. A 1-of-1 delivery record is not a rate, and scoring it as one would produce a confident-looking number built on almost nothing.

5.7 Peer-Relative Comparison (Secondary Signal)

Steps 5.1–5.5 apply one fixed bar to every supplier, which is intentional: leadership needs a consistent definition of Critical across the whole supplier base. But a fixed bar also flattens real differences between categories. Peer comparison adds a smaller, secondary signal: for each product category, it averages the Risk Score of scored suppliers, then flags a supplier as a category laggard when it sits more than a configurable gap (default 0.5 points) below that average, with at least one other scored peer to compare against. This never replaces the absolute Risk Tier. Comparing suppliers only within their own category can hide category-

wide risk, if every supplier in a category is structurally risky, they can all look average next to each other and nobody gets flagged. The absolute tier from Step 5.5 is what catches that; peer comparison only ever adds context on top of it.

6. System Architecture and Technology Stack

The application is built as a single self-contained React component with no external backend. This was a deliberate constraint: the tool needed to be shareable as one file and run entirely client-side.

Layer	Approach
UI framework	React (functional components and hooks)
Visualization	Hand-built SVG for the risk matrix, chosen over a charting library for full control over quadrant shading, bubble sizing by spend, and click-to-navigate behavior
Data import / export	PapaParse for CSV parsing and generation, with fuzzy column-header matching so imports tolerate real-world file variation
Iconography	lucide-react
Persistence	Browser-scoped key-value storage; autosave is debounced, and a two-step confirmation guards the reset-all action
Styling	Hand-written CSS with design tokens; no UI component library

7. Feature Overview

7.1 Register (Dashboard)

- A hero KPI (Spend at Risk) with a supporting stat strip, rather than a uniform grid of equally weighted cards.
- The reliability matrix: on-time rate against dependency score, bubble-sized by annual spend, colorable by Risk Tier, Category, or Region, click any point to open that supplier in the Ledger.
- Risk tier tally, a Watchlist of Critical/High suppliers, and a Category Laggards panel driven by the peer-comparison signal.
- A manual "Record Snapshot" action that logs every scored supplier's current standing for later trend comparison.

7.2 Supplier Ledger

- A fully editable data grid: every input field, inline, no separate edit mode.
- Sortable, filterable by risk tier, and searchable by name or location.
- Sticky identity columns so the supplier name stays visible while scrolling a wide table, plus a column show/hide control for a denser or more focused view.
- CSV import and export, with tolerant header matching for real-world file variation.
- Deletion is undo-able: a row disappears immediately but a toast offers Undo for several seconds rather than an instant, irreversible removal.
- Each row expands into a full calculation breakdown, showing every intermediate number that produced the final Risk Tier.

7.3 Settings

Every weight, threshold, multiplier, and guard described in Section 5 is editable here and recalculates the entire application immediately. A dedicated Data panel shows persistence status and provides a two-step "Reset all data" control.

7.4 Methodology

The full scoring methodology, including the MCDM comparison in Section 4.2 and the compounding-penalty rationale in Section 5.3, is documented inside the application itself and opened on demand from an unobtrusive trigger rather than occupying a permanent tab, so it supports the tool without competing with it for attention.

8. Design and User-Experience Evolution

The visual design went through several deliberate iterations rather than settling on a first pass:

- An early direction reused the dark, control-tower aesthetic of a prior, unrelated shipment-tracking project. This was intentionally abandoned in favor of a distinct identity: an “enterprise procurement ledger” look built around a parchment palette, institutional navy, and a brass accent for priority and authority elements.
- A confidentiality pass replaced every example supplier name and location with fully fictional alternatives after recognizing that an earlier, lightly anonymized set still implied an identifiable industry and geography pattern (see Section 9).
- A professionalization pass removed permanent, self-explanatory callout boxes in favor of on-demand tooltips, moved the Methodology content out of the primary navigation and

into an overlay, replaced a uniform grid of KPI cards with a clear data hierarchy, and reduced decorative use of color to the places where color actually carries meaning.

- A quality-of-life pass added sticky table columns, per-column visibility control, and undo-based row deletion in place of instant, irreversible actions.
- Two data-integrity issues found through direct testing, an on-time delivery count that could exceed total orders and produce a rate above 100%, and an annual-spend field that silently accepted negative values, were identified and fixed, with the corrected values now validated at both the input and calculation layers.

9. Data Privacy and Confidentiality Approach

- All example supplier names, locations, and industries in this prototype are invented. None correspond to any real company, and none are derived from the original internship dataset beyond matching its general statistical shape (risk-tier spread, sole-source exposure, thin order histories).
- The application has no backend server. Persisted data is stored locally in the browser session and is never transmitted elsewhere.
- Any compliance- or watchlist-style feature in the tool is explicitly illustrative and is not, and does not claim to be, a legal sanctions or denied-party screening.

10. Testing and Validation Approach

Because this tool has no build pipeline or hosted test runner, validation was performed directly against the compiled application before each change was delivered:

- Static parse verification with Babel ahead of every release, catching syntax errors before they reach the browser.
- Server-rendered smoke tests across every tab of the application to catch runtime errors that static parsing cannot.
- Interactive, simulated-browser testing (jsdom) for stateful behavior: the autosave-and-reload persistence cycle, snapshot history recording, undo-delete, column visibility toggling, modal open and close, and matrix-to-ledger navigation.
- Manual cross-checking of computed scores against the original spreadsheet's output values, confirming the reimplementation preserves the source methodology exactly.
- Targeted edge-case testing, including zero suppliers on file, on-time deliveries exceeding total orders, and negative spend entry, the last two of which surfaced the fixes described in Section 8.

11. Known Limitations and Future Work

Gap	Why It Was Left Out	Possible Next Step
No lead time, last-order-date, owner, or Incoterms fields	Deliberately scoped out to keep the initial field set defensible against real, thin procurement data	Add as an optional second tier of fields once core scoring is validated in use
No shareable PDF or print export	CSV export covers the data; a formatted report was out of scope for this pass	Add a print stylesheet or a generated summary export
No fleet-level "what changed" summary	Per-supplier trend exists; an aggregate roll-up does not yet	Summarize tier movement across all suppliers since the last snapshot
No bulk supplier actions	Single-row editing was sufficient for the current supplier count	Multi-select with bulk priority or category edits
No mobile-specific layout	Dense data grids are the core use case, prioritized for desktop	A condensed mobile view of the Register and Ledger tabs
Peer comparison can be masked category-wide	Inherent to any peer-relative signal; mitigated by keeping it secondary to the absolute tier	A category-level alert when an entire category trends downward together

12. Conclusion

The Supplier Reliability Matrix demonstrates a specific, defensible position: that a transparent, thinly-parameterized scoring model can outperform a more sophisticated one when the real constraint is data quality rather than analytical power. Every methodological choice in this document, what was included, what was left out, and what was fixed after testing, was made deliberately and is recorded here so it can be questioned. That is the intended contribution of this prototype: not just a working tool, but a documented, reviewable account of the reasoning behind it.

— End of document —